

Linear Algebra Final Review

1. Tell everything you can determine by looking at the reduced row echelon form of a matrix.

Finding a solution to a system of equations, rank, nullity, basis for column space, basis for null space, basis for row space, number of free variables, the weights in a linear combination, coordinate vectors, linear independence of columns, whether the columns span all of $\mathbb{R}^m$, is the matrix invertible, is a linear transformation T one-to-one, is it onto, is a vector in the range of T , find all vectors in the kernel of T .

2. Vector spaces.

What properties must a vector space have? List some properties of vector addition and scalar multiplication. How do you prove that something is a subspace? Give some examples? What is the dimension? What is a spanning set? What is a linearly independent set? What is a basis? How do you decide if a given set is linearly independent? How do you extend a given linearly independent set in $\mathbb{R}^m$ to form a basis for $\mathbb{R}^m$? How do you reduce a spanning set in $\mathbb{R}^m$ to form a basis for $\mathbb{R}^m$? How do you convert from abstract vector spaces to vectors in $\mathbb{R}^m$? What are the four fundamental subspaces associated with any $m \times n$ matrix?

3. Linear transformations.

Give an example. How do you prove a given function is a linear transformation? How do you find a matrix for T ? How are matrices for T related when you use different bases? What is the kernel of T ? What is the range of T ? How do you decide if a given vector is in the kernel of T ? What are similar matrices? How do you decide if T is invertible? What is the matrix for the inverse of T ?

4. Determinants and eigenvalues.

How do you calculate determinants? What is the effect of row operations on a determinant? How is the determinant affected by transpose, inverse, similarity? What is the connection between determinants and inverses? What can you say about the determinant of a product? What can you say about the determinant of an orthogonal matrix? What is the connection between determinants and eigenvalues? How do you find the eigenvalues of a matrix? How do you find a basis for the eigenspace for a given eigenvalue? What is the connection between eigenspaces and nullspaces?

5. Inner products.

What are the properties of dot products? How do you compute the length of a vector? How do you decide if two vectors are orthogonal? How do you construct a unit vector out of a given vector? What is the orthogonal complement of a vector space, W . How are orthogonal complements related to the four fundamental subspaces associated with any $m \times n$ matrix? How do you compute the projection of a vector v onto a subspace W if you have an orthogonal basis for W ? How do you convert a given basis into an orthogonal basis (Gram-Schmidt)? How do you find a least-squares solution to $Ax = b$.